

■ BASIC MODEL

$$\Rightarrow V(x) = (1-\delta)x + \delta \cdot [p \cdot V(x+fx) + q \cdot V(x(1-f))]]$$

where δ : the prob of casino will be open in the next period.

f : our betting fraction of money

p : the prob of winning a gamble.

$$\& \ q = 1-p$$

we set our seed as $V_0(x) = x$

$\Rightarrow$ Our way to plot $V_0(x), V_1(x) \dots$

① find the optimal betting fraction $\hat{f}$.

② use our $\hat{f}$ to get $V_t(x)$.

$\Rightarrow$ Realization

① Set-up:

- set winning prob $p = 0.8$

- for safety (guarantee a convergence),

$$\text{we need } \delta(2 \cdot p - q \cdot 0) < 1 \Rightarrow \delta < 0.625$$

and we set $\delta = 0.3$

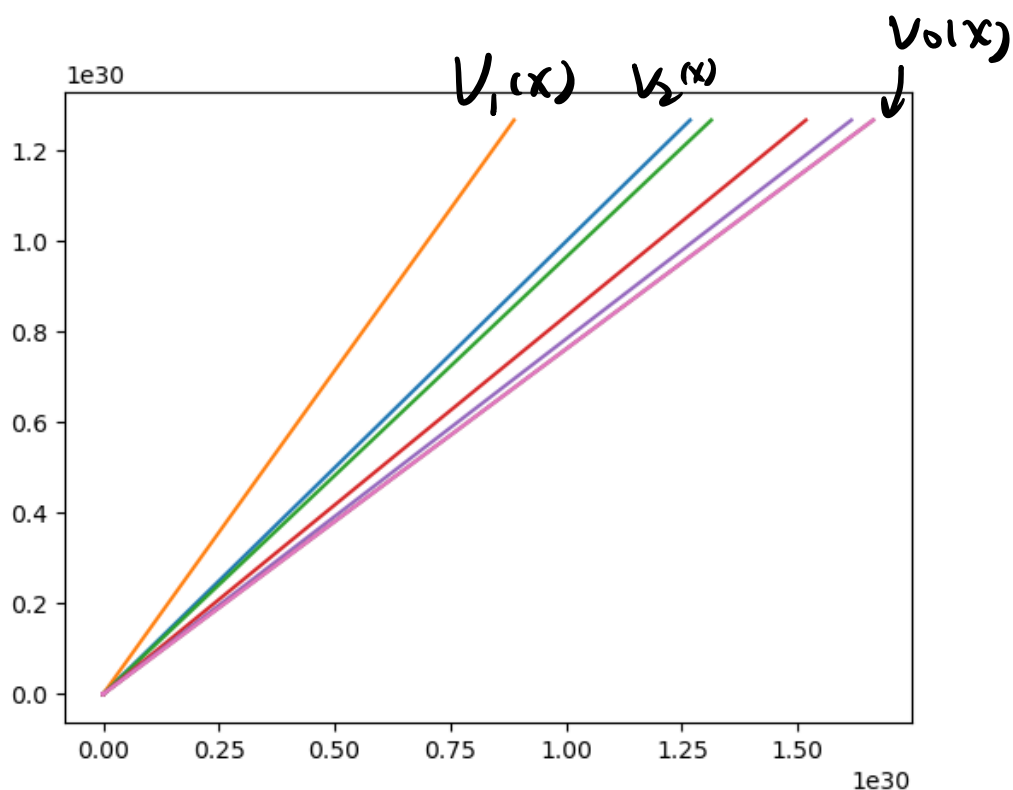
② Under this environment, we find the optimal betting fraction $\hat{f} = 1$.

p.s. we tried other winning probs & casino opening probs, but the result is bang-bang

and only depends on winning prob p .

explicitly: $\hat{f} = \begin{cases} 1 & \text{when } p > \frac{1}{2} \\ 0 & \text{when } p < \frac{1}{2} \end{cases}$

③ with betting fraction $f = 1$, we plot $V_t(x)$ where $t = 0, 1, 2, \dots$



NOTE:

- $V_t(x)$ are above 45° line ($V_0(x)$)
- $V_t(x)$ are straight lines

■ MODEL WITH GEOMETRIC UTILITY

$$\Rightarrow V(x) = (1-\delta) \cdot x^{1-\delta} + \delta \cdot [p \cdot V(x(1+f)) + q \cdot V(x(1-f))]$$

• Our revision here is to modify x to $x^{1-\delta}$.

$\Rightarrow$ REALIZATION:

• set-up: (we set the same numbers as the basic model)

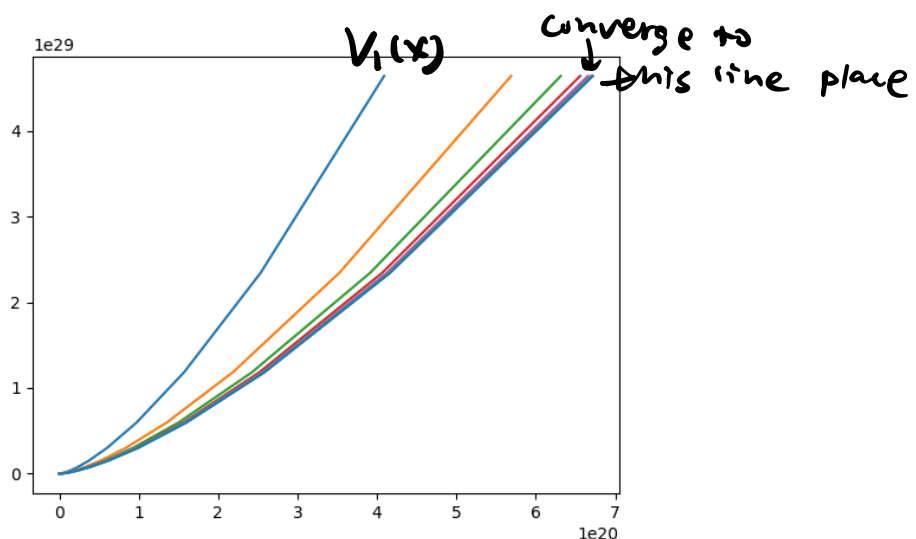
→ winning prob $p = 0.8$

→ casino opening prob $\delta = 0.3$

• find the optimal betting fraction

$$\hat{f} = 0.98$$

• Given $\hat{f} = 0.98$, we plot $V_t(x)$:



NOTE:

- the $V(x)$ lines are convex now
- They are still above the 45° line.

THE W-FUNCTION

⇒ The Derivation of w -function:

$$V(x) = (1-\delta) \cdot x + \delta \cdot [p \cdot V(x(1+f)) + q \cdot V(x(1-f))]$$

$$\text{Set } w(x) = \log(V(x)^{1-\delta})$$

$$= (1-\delta) \cdot \log V(x)$$

$$\text{Then } V(x) = \exp\left(\frac{w(x)}{1-\delta}\right)$$

$$e^{\frac{w(x)}{1-\delta}} = (1-\delta)x + \delta \left[p \cdot e^{\frac{w(x(1+f))}{1-\delta}} + q \cdot e^{\frac{w(x(1-f))}{1-\delta}} \right]$$

$$\frac{w(x)}{1-\delta} = \log \left\{ (1-\delta)x + \delta \left[p \cdot e^{\frac{w(x(1+f))}{1-\delta}} + q \cdot e^{\frac{w(x(1-f))}{1-\delta}} \right] \right\}$$

Then:

$$w(x) = (1-\delta) \cdot \log \left\{ (1-\delta)x + \delta \left[p \cdot e^{\frac{w(x(1+f))}{1-\delta}} + q \cdot e^{\frac{w(x(1-f))}{1-\delta}} \right] \right\}$$

And we set seed as

$$w_0(x) = (1-\delta) \cdot \log(x)$$

$$\text{Since } V_0(x) = x \quad \& \quad w(x) = (1-\delta) \cdot \log V(x)$$

⇒ Realization of Plotting $V_t(x)$

• The best betting fraction $\hat{f} = 1$

